

DRSPLUS RETINOGRAPHER IN PATIENTS WITH DIABETES

A Comparison Study by Consensus Grading, Reading Center, and Artificial Intelligence

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Purpose: To evaluate the gradable rate of the retinal images acquired with DRSplus retinographer in patients with diabetes and to estimate the diabetic retinopathy (DR) severity, comparing different methods of analysis and grading.

Methods: Prospective, cross-sectional, observational study. A mosaic of overlapped retinal images in nonmydriatic condition was acquired, evaluating the gradable images rate at both image and eye-levels, from the human-study team (consensus grading), Reading Center, and an Artificial Intelligence (AI) system, respectively. The DR severity was graded by the Reading Center and the AI.

Results: A total of 844 images of 422 eyes from 224 patients were included. The gradable rate was 87.4%, 97.4%, and 96.9% at image-level and was 81.3%, 97.4%, and 95.0% at eye-level, according to the consensus grading, the Reading Center, and the AI, respectively. Artificial intelligence sensitivity for detecting referable DR was 97.3%, the specificity was 80.4%, and the accuracy was 86.6%.

Conclusion: DRSplus retinographer allowed to image the retina in nonmydriatic condition in more than 80% of eyes, with a higher gradable rate with Reading Center and AI in comparison with consensus grading. The estimation of the DR severity showed a high sensitivity for referral cases for AI system, indicating the potentialities of telemedicine-based tool for the DR screening.

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The diabetic retinopathy (DR) is one of the major complication of diabetes, with an overall prevalence of 35% representing an important cause of blindness in people at working age throughout the world.^{1,2}

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Several aspects need to be considered, such as the high-level and widespread diffusion of care for DR and diabetic macular edema, the impact of referable diabetic eye disease at the time of diabetes' diagnosis, and the significant proportion of vision loss avoidable through early detection and timely treatment.^{1,3,4}

An ever-increasing interest on systematic screening program for retinopathy detection in people with diabetes has long been endorsed as an important public health intervention and in continue evolution.^{1,4}

Screening for sight-threatening retinopathy has been shown to be both, clinical and cost-effective.¹ Despite these evidences, only approximately half of all diabetic patients underwent DR screening and, consequently, only a minor part of patients could effectively access to the treatment if needed.^{5–7}

The screening programs allow to distinguish between referable or nonreferable patients, to schedule the visits and the timing for follow-up.

In this context, the implementation of programs using telemedicine is an important target for DR screening.^{3,4,7,8}

Several nonmydriatic (NM) fundus camera devices have been developed to noninvasively capture retinal imaging allowing safe and cost-effective screening programs, ultimately resulting in finding fundus pathologies also beyond DR, very similarly to fundus examination in an outpatient ophthalmic setting.⁹

The gold standard for the detection and grading of DR was the 7 standard 35-mm stereoscopic color fundus photographs (7F-ETDRS); however, with the advent of new fundus camera systems, it can be replaced by three color 45° NM fundus fields with high sensitivity and specificity to determine critical levels of DR and diabetic macular edema for prompt specialist referral.⁷

A previous study using DRSplus device showed that NM montage widefield images were compared favorably with dilated fundus examination in defining DR severity, reducing the time acquisition.¹⁰

The aim of the present study was to evaluate the gradable rate of the retinal images acquired with DRSplus retinographer in patients with diabetes, comparing human-study team (consensus grading), Reading Center (RC), and Artificial Intelligence (AI), and to estimate DR severity, comparing different methods of analysis and grading.

Methods

Study Design

Prospective, cross-sectional, observational, monocentric study conducted at IRCCS-Fondazione Bietti in Rome, Italy. The study received Institutional Review Board approval (Comitato Etico Centrale IRCCS Lazio, prot. N106/20/FB) and followed the tenets of the Declaration of Helsinki. Written informed consent was obtained from all enrolled patients.

Inclusion and Exclusion Criteria

Inclusion criteria were 1) age 18 to 90 years, 2) diagnosis of Type 1 or Type 2 diabetes (T1DM or T2DM) with or without DR, and 3) treatment-naïve or previously treated patients (both, with laser photocoagulation and/or intravitreal injections). Exclusion criteria were 1) other retinal pathologies than DR, 2) media opacities, and 3) refractive errors (i.e., myopia < -6.00 D spherical equivalent and astigmatism (cylinder) > ±3.00 D).

Study Procedure

After signing informed consent, all patients were imaged using DRSplus NM fundus camera (Centervue

SpA, Padova, Italy). All the procedures were performed according to the good clinical practice.

The DRSplus device is a confocal system that captures 45° × 40°, 10-megapixel retinal images in an automated fashion through a nonmydriatic pupil. It also allows to capture montage wider images (up to 83 × 78°) with a mosaic of partially overlapped images. The light source is a broad spectrum white-light Light Emitting Diode (LED) (440–650 nm).

Images were acquired by trained ophthalmology assistants, who obtained a mosaic of partially overlapped retinal images covering 63° of the retina (from central to nasal) mostly in nonmydriatic condition:

1. One color image of the central field
2. One color image of the nasal field

A subgroup of patients received a pupil dilation, and the images were acquired in mydriatic condition.

Notably, only one retake of the image was performed if a blink occurred during the acquisition.

Primary Outcomes

The primary outcomes were 1) the evaluation of the gradable rate, defined as the percentage of gradable images over the total number of acquired DRSplus images, and 2) the evaluation of DR severity grading.

Sample Size Calculation

The study variable of the primary outcomes was qualitative: percentage of gradable images over total acquired images. Previous data showed a mean gradable rate approximately 85%.^{10–14} Considering a clinically significant effect size of at least 5%, the number of eyes of different subject needed to prove an increase from 85% to 90% was 402, to assure the level of significance 5% and power 85% (tested via two side binomial test for proportion). Sample size calculation was performed using G*Power version 3.1.9.7.

Data Collection and Statistical Analysis

All data collected were anonymized and stored in a database.

The study sample characteristics were summarized with descriptive statistics including means, SD, medians, and ranges. Categorical variables (e.g., diabetes type, gender stratification, mydriatic/nonmydriatic information, and DR severity grading) were summarized with frequencies and percentages.

The percentage of gradable images over the total number of acquired DRSplus images was calculated (“gradable rate”) by the human-study team (consensus grading), the RC, and the AI system, respectively (see

below). To estimate the agreement, the Cohen kappa statistic was used. Such method was used to evaluate the agreement for both gradeability (binary outcome) and DR severity grading (ordinal variables) evaluations. To penalize disagreements in DR grading seriousness, a variation of the Cohen kappa (called weighted kappa with linear weights) was used for DR grading only.

All data were analyzed using Microsoft Excel for Microsoft 365 (version 2311) and R (R Foundation for statistical computing, Vienna, Austria) software.

Gradable Image Rate Analysis

The gradable image rate was defined and analyzed in a masked fashion in three different ways:

1. By four experienced ophthalmologists (human-study team), expressing a consensus grading.
2. By an external RC (InHealth Intelligence, Universal House, Middlewich) (human-ground truth).
3. By an AI system, RetCAD by Thirona Retina B.V. (Nijmegen, The Netherlands). RetCAD is a commercially available, Class IIa CE-marked medical device software driven by AI that detects DR in color fundus images.

The gradable image rate was expressed as “gradable” or “ungradable” at two different levels:

1. Image-level (evaluation of a single retinal image)
2. Eye-level (evaluation of both, central and nasal images of the retinal montage).

In particular, the images were reviewed by four experienced ophthalmologists (human-study team) who provided a gradeability label (“gradable” or “ungradable”) based on their consensus (consensus grading). The “ungradable” label was assigned when inadequate image quality prevented determining the presence of a characteristic being graded. The gradable rates at image-level were then calculated; if at least one image (image-level) was judged as “ungradable,” the whole eye (eye-level) was defined “ungradable.”

The gradable image rate analysis of the RC consisted of a heuristic model for which all the images of an eye (image-level) shared the quality score provided at eye-level.

The gradable image rate analysis of the RetCAD AI system assigned a quality score [0–100] for each image, defined as “gradable” a score ≥ 25 . If at least one image (image-level) was judged as “ungradable,” the whole eye (eye-level) was defined “ungradable.”

The RC was considered as reference standard.

Agreement in quality evaluations between consensus grading (human-study team) versus RC, consensus

grading versus AI system, and RC versus AI system was evaluated using the Cohen Kappa method.

Diabetic Retinopathy Severity Grading

The DR severity was graded by the following:

1. The external RC (InHealth Intelligence, Universal House, Middlewich), which provided DR grades according to the criteria of the English NHS Diabetic Eye Screening Programme (<https://www.gov.uk/government/publications/diabetic-eye-screening-retinal-image-grading-criteria/features-based-grading-outcomes-guidance>; <https://www.gov.uk/government/publications/diabetic-eye-screening-retinal-image-grading-criteria/nhs-diabetic-eye-screening-programme-grading-definitions-for-referable-disease-start-date-october-01>).^{15,16}
2. An AI system, RetCAD by Thirona Retina B.V. (Nijmegen, The Netherlands), which used the International Clinical Diabetic Retinopathy (ICDR) Disease severity Scale.¹⁷

The RC staging was used as reference standard. The DR staging was performed at eye-level to obtain a paradigm of the patients needing a specialist referral.

A match between the two scales (National Health Service (NHS) and ICDR) was performed according to Meredith et al,¹⁸ and is reported in the Table 1.

Nonreferable retinopathy was defined as no or mild DR, while referable retinopathy was defined as moderate DR or worse.^{6,19} Confusion matrices were derived to compare the DR scores assigned by the AI system and the RC at eye-level, with sensitivity, specificity, accuracy, and positive predictive value (PPV) metrics together with their 95% confidence intervals (CI). All the analysis were conducted considering the overall population and, furthermore, a subanalysis for right and left eye (OD and OS) was conducted.

Finally, the Receiver Operating Characteristics curve was derived at eye-level for the AI system.

Results

The study included 844 images of 422 eyes from 224 patients with diabetes (58.5% males, mean age 66.61 ± 12.60 years, 92.9% with T2DM). Approximately 91.2% of eyes were imaged in nonmydriatic conditions (mean pupil size = $3.25 \text{ mm} \pm 0.79 \text{ SD}$), while 8.8% of eyes were pharmacologically dilated (mean pupil size = $5.27 \text{ mm} \pm 0.83 \text{ SD}$). The study sample characteristics are summarized in Table 2.

Table 1. English NHS Diabetic Eye Screening Program and ICDR Severity Scale Match

ICDR Severity Scale/Corresponding AI System's Scores	English NHS Diabetic Eye Screening Program
No DR/Stage 0: (0–25)	R0: No retinopathy
Mild DR/Stage 1: (25–50)	R1: Background retinopathy
Moderate DR/Stage 2: (50–67)	R2: Preproliferative retinopathy
Severe DR/Stage 3: (67–84)	R2: Preproliferative retinopathy
Proliferative DR/Stage 4: (84–100]	R3: Proliferative retinopathy

Gradable Rate Analysis Results

Gradable rate analysis results are reported in Table 3. The gradable rate at image-level was 87.4%, 97.4%, and 96.9% according to the consensus grading, the RC, and the AI system, respectively. At eye-level, the gradable rate was 81.3%, 97.4%, and 95.0%, according to the consensus grading, the RC, and the AI system, respectively. The differences in proportions were respectively 0.47%, 95% CI: [0.01%–0.94%] for AI versus RC, 9.48%, 95% CI: [7.50%–11.45%] for AI versus consensus grading, and 9.95%, 95% CI: [7.93%–11.97%] for RC versus consensus grading.

At image-level, fair agreement in quality evaluations was found between consensus grading versus RC (Cohen kappa coefficient $k = 0.27$, 95% CI: [0.17–0.36]), AI system versus consensus grading ($k = 0.35$, 95% CI: [0.24–0.45]), and AI system versus RC ($k = 0.31$, 95% CI: [0.14–0.49]). A similar fair agreement was found in the subanalysis for OD and OS.

Diabetic Retinopathy Severity Grading Results

After excluding 11 eyes of 11 patients judged as ungradable by the RC, a total of 411 eyes of 213 patients was considered for DR severity analysis. Confusion matrices comparing AI system and RC

Table 2. Study Sample Characteristics Including DR Grading Distribution According to RC

	N	%
N total patients	224	100.0
N males	131	58.5
N females	93	41.5
Diabetes Type: 1	16	7.1
Diabetes Type: 2	208	92.9
N total eyes	422	100.0
N mydriasis	37	8.8
N miosis	385	91.2
DR grading—R0	143	33.9
DR grading—R1	117	27.7
DR grading—R2	79	18.7
DR grading—R3	72	17.1
DR grading—U	11	2.6

U, ungradable.

DR scores at eye-level are reported in Table 4. Confusion matrices for nonreferable and referable DR are reported in Table 5. Data on the confusion matrices for gradable/ungradable images and eyes between RC, AI system, and Consensus Grading were reported in the **Supplemental Digital Content 1** (see Supplementary tables 1a and 1b, <http://links.lww.com/IAE/C765>).

At the eye-level, the AI system sensitivity for detecting referable DR was 97.3%, 95% CI: [94.74%–99.86%], the specificity was 80.4%, 95% CI: [75.57%–85.23%], and the accuracy was 86.6%, 95% CI: [83.31%–89.89%]. The PPV was 74.24%, 95% CI: [68.15%–80.33%].

Four eyes (0.97%) of two patients with referable DR for RC were deemed to have nonreferable DR by the AI system. However, 51 eyes (12.4%) of 27 patients with nonreferable DR for the RC were misclassified as referable DR by the AI system.

The subanalysis for OD and OS showed similar performance indices for detecting referable DR with a sensitivity of 97.30% 95% CI: [93.60%–100%] for OD and 97.40% 95% CI: [93.85%–100%] for OS; specificity of 79.56% 95% CI: [72.81%–86.31%] for OD and 81.30% 95% CI: [74.41%–88.19%] for OS; accuracy of 85.78% 95% CI: [81.07%–90.49%] for OD and 87.50% 95% CI: [82.92%–92.08%] for OS, and PPV of 72.00% 95% CI: [63.20%–80.80%] for OD and 76.53% 95% CI: [68.14%–84.92%] for OS.

Substantial agreement was found between the AI system and the RC (weighted Cohen kappa $k_w = 0.62$, 95% CI: 0.57–0.66).

The subanalysis for OD and OS showed similar agreement between AI system and RC (OD: $k_w = 0.61$, 95% CI: 0.55–0.67; OS: $k_w = 0.62$, 95% CI: 0.56–0.68).

The receiver operating characteristics curve was derived by plotting sensitivity against specificity at various threshold values for AI system's DR referability (Figure 1). The area under the receiver operating characteristic curve was 0.979, 95% CI: [0.0968–0.990].

Data on the analysis in mydriatic versus nonmydriatic conditions to assess the gradeability and the DR severity were reported in the Tables 6 and 7,

Table 3. Gradable Rate Analysis at Image and Eye-Level

Gradable Rate Analysis	Total	Ungradable	Gradable
	N [%]	N [%]	N [%; 95% CI]
Image level			
AI system	844 [100%]	26 [3.1%]	818 [96.9%; 95.7%–98.1%]
Reading center	844 [100%]	22 [2.6%]	822 [97.4%; 96.3%–98.5%]
Consensus grading	844 [100%]	106 [12.6%]	738 [87.4%; 85%–89.8%]
Eye level			
AI system	422 [100%]	21 [5.0%]	401 [95.0%; 92.9%–97.2%]
Reading center	422 [100%]	11 [2.6%]	411 [97.4%; 95.9%–98.9%]
Consensus grading	422 [100%]	79 [18.7%]	343 [81.3%; 77.2%–85.4%]

respectively. In myosis condition, the AI and RC assessment of nonreferable and referable DR showed a sensitivity of 98.50%, 95% CI: [96.43%–100.00%], a specificity of 80.0%, 95% CI: [74.83%–85.17%], an accuracy of 86.78%, 95% CI: [83.29%–90.26%], and a PPV of 74.01%, 95% CI: [67.55%–80.47%]; in mydriasis condition, the performance showed a sensitivity of 92.86%, 95% CI: [79.37%–100.00%], a specificity of 94.44%, 95% CI: [83.86%–100.00%], an accuracy of 93.75%, 95% CI: [85.36%–100.00%], and a PPV of 92.86%, 95% CI: [79.37%–100.00%].

Subanalyses for OD and OS were reported in the **Supplemental Digital Contents 2, 3, 4, 5, 6, 7**; Supplementary tables (2a and 2b, 3a and 3b, 4a and 4b, 5a and 5b, 6a and 6b, 7a and 7b, <http://links.lww.com/IAE/C766>, <http://links.lww.com/IAE/C767>, <http://links.lww.com/IAE/C768>, <http://links.lww.com/IAE/C769>, <http://links.lww.com/IAE/C770>, and <http://links.lww.com/IAE/C771>).

Discussion

In this study, we aimed to evaluate the gradable rate of retinal images acquired with the DRSplus scanning ophthalmoscope in patients with T1DM and T2DM, comparing human-study team (consensus grading), RC, and AI (RetCAD), and to estimate the DR severity comparing different methods of analysis and grading. The DRSplus retinographer allows to obtain confocal

images of the retina, even without pharmacologic dilation.^{9,10}

Regarding the first part of our outcome, the gradable rate was evaluated on 844 images classified as “gradable” or “ungradable” by human-study team (consensus grading), RC, and AI system, which respectively showed a high percentage of gradable images (87.4%, 97.4%, and 96.9%, respectively), with fair agreement in quality evaluations between different modalities.

The analysis of the gradable rate at eye-level showed a lower percentage of gradeability expressed by the consensus grading when compared with RC and AI rates (81.3%, 97.4%, and 95.0%, respectively). This result is mostly related to the difference in the determination of the gradeability between the three methods of analysis, as the consensus grading needed plenary consensus among experienced ophthalmologists to obtain the gradable attribution. In this way, the consensus grading was more challenging for final attribution and a higher number of images were defined as ungradable.

On the other side, the analysis of AI system gradeability rate showed a high correspondence with the consensus grading and RC, but for all the systems, we found a reduction of gradable image rate in myosis condition (80.3%, 89.3%, and 88.5% at eye-level for human study team, RC, and AI system, respectively). The gradeability rate in myosis condition could be probably because of the presence of some artifacts, such as occurred in case of very low pupil size,

Table 4. Confusion Matrices Comparing AI System and Reading Center DR Severity Scores at Eye-Level

	AI System (Eye-Level)				Total
	0	1	2, 3	4	
Reading center (eye-level)					
R0	114	27	2	0	143
R1	34	34	49	0	117
R2	1	1	62	15	79
R3	0	2	51	19	72
Total	149	64	164	34	411

Table 5. Confusion Matrices for Nonreferable and Referable DR

	AI System (Eye-Level)		Total
	Nonreferable DR	Referable DR	
Reading center (eye-level)			
Nonreferable DR	209	51	260
Referable DR	4	147	151
Total	213	198	411

eyelashes, or age-related postural difficulty that might have affected the image acquisition.

The quality of digital images for assessing and managing DR represents an important issue, in fact, low quality of digital images leads to potential loss of patients that are in need of a referral.¹² A recent study using DRSplus device reported a very low percentage of images (7.5%) with poor quality captured in NM conditions.¹⁰ In the mentioned study, patients with T1DM and T2DM performed a NM fundus imaging and a dilated fundus examination; interestingly, the

agreement between DR severity evaluated by indirect dilated fundus examination and by DRSplus in NM conditions was 97.3%. The results of this first study indicated that NM montage widefield images obtained with the DRSplus device have an elevated agreement with indirect dilated ophthalmoscopy.¹⁰

The previous results are in line with the results from the present analysis in which a high agreement between RC and AI system has been found to evaluate referable DR and the AI system showed a 97.3% of sensitivity for detecting referable cases. In our study,

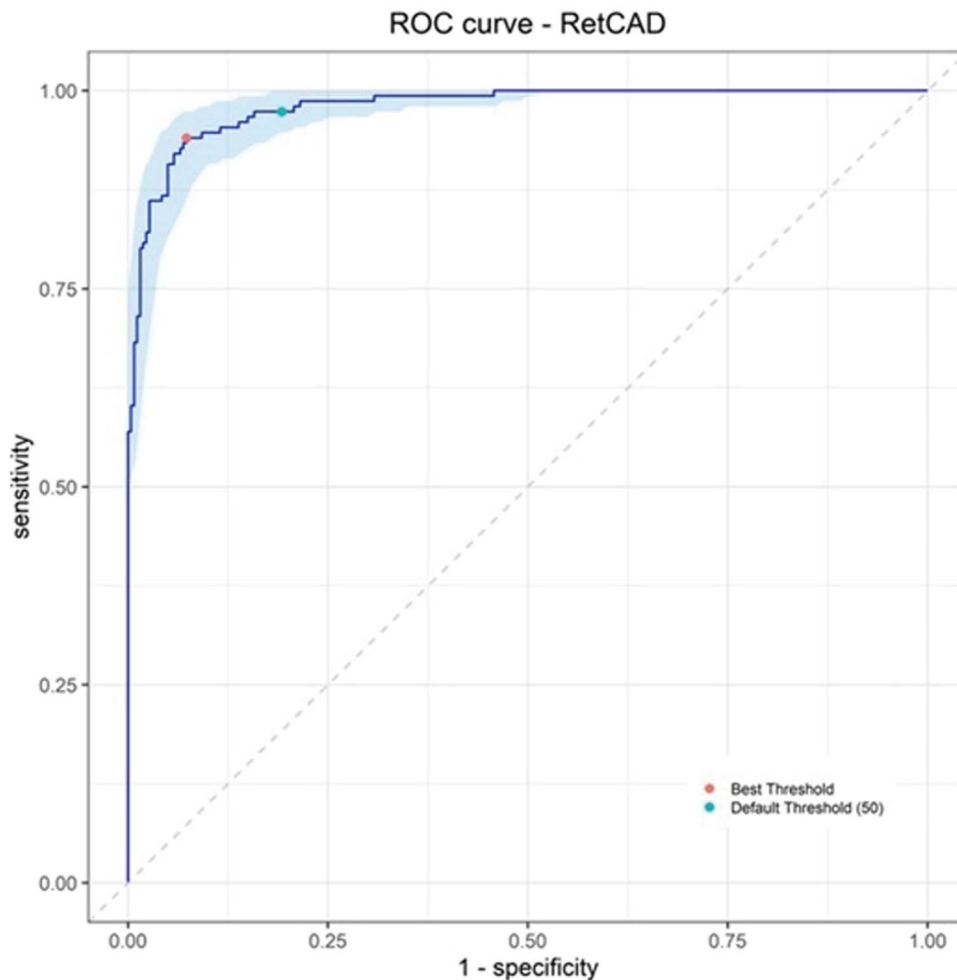


Fig. 1. The green dot represents the default threshold adopted by the AI system. In addition, for this specific dataset, we calculated the best threshold (red dot—point closest to the upper-left corner of the graph), corresponding to a 94% sensitivity and 92.69% specificity.

Table 6. Analysis Mydriatic Versus Nonmydriatic: Number and Percentage of Gradable Data Collected

Gradable Rate Comparison Myosis vs. Mydriasis	N [%] Total	N [%] Myosis Myosis	N [%] Myosis Gradable	N [%] Myosis Ungradable	N [%] Mydriasis	N [%] Mydriasis Gradable	N [%] Mydriasis Ungradable
Image level							
AI system	844 [100%]	770 [91.2%]	747 [97%]	23 [3%]	74 [8.8%]	71 [95.9%]	3 [4.1%]
Reading center	844 [100%]	770 [91.2%]	754 [97.9%]	16 [2.1%]	74 [8.8%]	68 [91.9%]	6 [8.1%]
Consensus grading	844 [100%]	770 [91.2%]	678 [88.1%]	92 [11.9%]	74 [8.8%]	60 [81.1%]	14 [18.9%]
Eye level							
AI system	422 [100%]	385 [91.2%]	367 [95.3%]	18 [4.7%]	37 [8.8%]	34 [91.9%]	3 [8.1%]
Reading center	422 [100%]	385 [91.2%]	377 [97.9%]	8 [2.1%]	37 [8.8%]	34 [91.9%]	3 [8.1%]
Consensus grading	422 [100%]	385 [91.2%]	315 [81.8%]	70 [18.2%]	37 [8.8%]	28 [75.7%]	9 [24.3%]

the RC has been used as the reference standard for assessing the severity of DR on NM images captured by DRSplus device. In particular, the analysis of eyes with no or mild DR (R0 and R1 in NHS classification matched as level 0 and 1 in ICDR classification) showed an excellent matching between RC and AI. Considering the advanced forms of DR, some cases of proliferative DR (R3 level in NHS) have been misclassified as severe DR from the AI systems (level 2,3 in ICDR), instead of level 4. This mismatching did not negatively influence the referral rate, in fact all of these stages corresponded to levels of DR needed referral.^{6,19}

Our study presented some limitations such as the lack of comparison to a gold standard assessment for fundus images evaluation; the absence of a head-to-head comparison between myotic and mydriatic images to test whether the pupil size significantly affected the gradable rate; the different way of image grading by the consensus group (with more stringent criteria of gradable attribution), by the RC (which used

a heuristic model), and by the AI system, respectively; the different DR severity scales used by the RC and the AI system, which have been matched following Meredith at al¹⁸ criteria; and furthermore, in the acquisition procedure, only one retake was permitted in case of blinks, potentially reducing the number of well-taken gradable images. Another limitation was the light overestimation of referral for the AI system (as reported in the Table 5) in particular for the advanced forms of DR, which needed for further amelioration of this DR screening system. Interestingly, the subanalysis per single eye (OD and OS) showed similar agreement and performance's indices than the analysis on the overall population, for both, the gradeability and the DR severity.

Our results agreed with those reported by Borrelli et al¹⁰ confirming the rapid and favorable detection of DR by DRSplus device, opening an interesting scenario on the potential of a telemedicine-based tool for the DR screening for this device.

Table 7. Confusion Matrices Comparing Reading Center and AI System Assessment of Nonreferable and Referable DR at Eye-Level, Stratified by Myosis and Mydriasis

	AI System (Eye-Level)		Total
	Nonreferable DR	Referable DR	
Myosis			
Reading center (eye-level)			
Nonreferable DR	184	46	230
Referable DR	2	131	133
Total	186	177	363
Mydriasis			
Reading center (eye-level)			
Nonreferable DR	17	1	18
Referable DR	1	13	14
Total	18	14	32

An automated imaging system powered by AI, such as the one used in this study, might reduce the burden for ophthalmologists, and in addition, this solution could be integrated in a more complex and widespread system able to reach patients based in rural areas.

The AI DR tool can assist the clinician, to detect early signs of DR optimizing treatments.

In the future, fully automated AI-based grading systems could support to decide whether a patient requires referral, or not, and could potentially be much cheaper than having ophthalmologists conduct screening.²⁰

In conclusion, the results of the present study showed that the DRSplus device allowed to collect retinal images of diabetic patients in nonmydriatic condition, with a high gradable image rate both for the consensus grading and the AI system. The estimation of the DR severity showed a high sensitivity for referral cases for AI system, indicating the potentialities of a telemedicine-based tool for the DR screening.

Key words: diabetic retinopathy, gradeability analysis, nonmydriatic fundus camera.

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References

1. Lawrenson JG, Bourmpaki E, Bunce C, et al. Trends in diabetic retinopathy screening attendance and associations with vision impairment attributable to diabetes in a large nationwide cohort. *Diabet Med* 2021;38:e14425.
2. Yau JW, Rogers SL, Kawasaki R, et al. Global prevalence and major risk factors of diabetic retinopathy. *Diabetes Care* 2012; 35:556–564.
3. GBD 2019 Blindness and Vision Impairment Collaborators. Vision Loss Expert Group of the Global Burden of Disease Study. Causes of blindness and vision impairment in 2020 and trends over 30 years, and prevalence of avoidable blindness in relation to VISION 2020: the right to sight: an analysis for the Global Burden of Disease Study. *Lancet Glob Health* 2021;9:e144–e160.
4. Amoaku WM, Ghanchi F, Bailey C, et al. Diabetic retinopathy and diabetic macular oedema pathways and management: UK Consensus Working Group. *Eye (Lond)* 2020;34:1–51.
5. McGlynn EA, Asch SM, Adams J, et al. The quality of health care delivered to adults in the United States. *N Engl J Med* 2003;348:2635–2645.
6. Vujosevic S, Midea E. Diabetic retinopathy in Italy: epidemiology data and telemedicine screening programs. *J Diabetes Res* 2016;2016:3627465.
7. Vujosevic S, Benetti E, Massignan F, et al. Screening for diabetic retinopathy: 1 and 3 nonmydriatic 45-degree digital fundus photographs vs 7 standard early treatment diabetic retinopathy study fields. *Am J Ophthalmol* 2009;148:111–118.
8. Cavallerano AA, Cavallerano JD, Katalinic P, et al. A telemedicine program for diabetic retinopathy in a veterans affairs medical center-the joslin vision network eye health care model. *Am J Ophthalmol* 2005;139:597–604.
9. Mastropasqua L, Perilli R, D'Aloisio R, et al. Why miss the chance? Incidental findings while telescreening for diabetic retinopathy. *Ophthalmic Epidemiol* 2020;27:237–245.
10. Borrelli E, Querques L, Lattanzio R, et al. Nonmydriatic wide-field retinal imaging with an automatic white LED confocal imaging system compared with dilated ophthalmoscopy in screening for diabetic retinopathy. *Acta Diabetol* 2020;57: 1043–1047.
11. Gosheva M, Klameth C, Norrenberg L, et al. Quality and learning curve of handheld versus stand-alone non-mydriatic cameras. *Clin Ophthalmol* 2017;11:1601–1606.
12. Kim HM, Lowery JC, Kurtz R. Accuracy of digital images for assessing diabetic retinopathy. *J Diabetes Sci Technol* 2007;1: 531–539.
13. Porta M, Boscia F, Lanzetta P, et al. Systematic screening of retinopathy in diabetes (REaD project): an Italian implementation campaign. *Eur J Ophthalmol* 2017;27:179–184.
14. Teismann N, Neilson J, Keenan J. Quality and feasibility of automated digital retinal imaging in the emergency department. *J Emerg Med* 2020;58:18–24.
15. Harding S, Greenwood R, Aldington S, et al. Grading and disease management in national screening for diabetic retinopathy in England and Wales. *Diabet Med* 2003;20:965–971.
16. Scanlon PH, Stratton IM, Leese GP, et al. Screening attendance, age group and diabetic retinopathy level at first screen. *Diabet Med* 2016;33:904–911.
17. Wilkinson CP, Ferris FL, Klein RE, et al. Proposed international clinical diabetic retinopathy and diabetic macular edema disease severity scales. *Ophthalmology* 2003;110:1677–1682.
18. Meredith S, van Grinsven M, Engelberts J, et al. Performance of an artificial intelligence automated system for diabetic eye screening in a large English population. *Diabet Med* 2023;40: e15055.
19. Scanlon PH, Aldington SJ, Leal J, et al. Development of a cost-effectiveness model for optimisation of the screening interval in diabetic retinopathy screening. *Health Technol Assess* 2015; 19:1–116.
20. Padhy SK, Takkar B, Chawla R, Kumar A. Artificial intelligence in diabetic retinopathy: a natural step to the future. *Indian J Ophthalmol* 2019;67:1004–1009.